

BROKEN BOW, NE, USA

WMO: 725555

Lat: 41.433N

Lon: 99.633W

Elev: 771

StdP: 92.40

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 94946

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-17.1	-23.4	0.5	-19.0	-21.1	0.6	-16.2	15.7	-0.2	14.4	-1.0	3.4	340	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	34.7	21.4	32.7	21.7	31.2	21.4	24.4	30.5	23.5	29.4	22.6	28.5	5.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	18.9	27.7	21.7	18.0	27.1	20.9	17.1	26.4	78.0	30.7	74.2	29.3	70.6	28.3	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.7	11.8	10.5	DB	-24.5	37.1	2.3	2.0	-26.2	38.6	-27.6	39.7	-28.9	40.8	-30.6	42.3	(4)
(5)			WB	-24.8	26.1	2.3	1.2	-26.4	27.0	-27.7	27.7	-29.0	28.3	-30.7	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-3.1	-2.8	3.8	9.1	14.8	20.7	23.7	22.1	17.5	9.9	3.1	-2.8	(6)	
	DBStd	10.96	6.50	6.75	6.13	5.40	4.83	3.81	3.02	3.11	4.68	5.17	5.89	6.19	(7)	
	HDD10.0	1742	407	358	207	78	13	0	0	0	3	64	216	397	(8)	
	HDD18.3	3595	664	591	452	279	131	20	3	6	71	265	458	656	(9)	
	CDD10.0	1639	0	1	13	51	161	320	423	374	228	61	8	0	(10)	
	CDD18.3	452	0	0	1	2	21	90	168	122	46	3	0	0	(11)	
	CDH23.3	5103	0	0	17	67	297	994	1823	1240	573	86	6	0	(12)	
	CDH26.7	2075	0	0	3	14	94	380	838	510	211	24	1	0	(13)	
(14) Wind	WSAvg	4.4	4.5	4.6	4.9	5.5	5.0	4.4	3.5	3.4	3.9	4.2	4.3	4.4	(14)	
(15) Precipitation	PrecAvg	573	10	13	34	58	85	99	87	63	54	33	22	10	(15)	
	PrecMax	894	41	44	148	153	185	262	230	199	153	91	71	27	(16)	
	PrecMin	287	0	0	0	5	19	25	6	3	6	0	0	0	(17)	
	PrecStd	105	10	11	28	35	44	59	49	38	40	25	19	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	18.7	25.8	28.7	32.5	35.9	37.5	36.2	33.8	30.1	23.7	16.0	(19)	
		MCWB	7.2	9.2	12.8	15.6	17.8	20.8	21.9	21.6	19.5	16.4	11.7	7.2	(20)	
	2%	DB	12.1	14.0	21.2	25.3	29.2	32.8	35.2	33.5	31.6	25.6	19.9	12.0	(21)	
		MCWB	5.1	6.7	10.7	14.1	18.0	21.3	21.6	21.7	20.2	14.9	10.2	5.3	(22)	
	5%	DB	8.8	10.6	17.7	22.3	26.8	30.9	33.0	31.5	29.0	22.6	16.3	8.7	(23)	
		MCWB	3.5	4.8	9.9	12.9	17.2	21.1	22.5	21.9	19.4	13.3	8.6	3.7	(24)	
	10%	DB	6.1	7.3	14.3	19.2	24.0	28.6	31.4	29.5	26.9	19.4	12.7	6.1	(25)	
		MCWB	2.1	3.2	7.9	11.5	16.0	20.1	22.3	21.7	18.0	11.9	6.9	1.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	10.4	15.1	16.8	21.1	24.5	25.8	25.1	23.1	18.0	13.1	8.4	(27)	
		MCDB	16.0	16.8	19.6	24.6	27.6	30.9	31.9	30.4	29.0	24.9	20.9	13.9	(28)	
	2%	WB	5.4	7.5	12.7	15.2	19.6	23.0	24.7	24.0	21.6	16.3	10.9	5.5	(29)	
		MCDB	11.5	13.3	18.5	22.5	26.4	29.1	31.1	30.0	28.2	22.2	17.4	11.4	(30)	
	5%	WB	3.8	5.3	10.3	13.7	18.2	22.1	23.9	23.0	20.5	14.7	9.1	3.8	(31)	
		MCDB	8.3	10.6	17.3	20.9	24.8	28.8	30.0	28.5	26.9	20.5	15.8	8.2	(32)	
	10%	WB	2.3	3.3	8.1	12.1	16.9	21.0	23.0	22.1	19.4	12.9	7.0	2.1	(33)	
		MCDB	5.8	7.1	14.4	18.2	22.7	27.4	28.9	27.7	25.0	18.5	12.8	5.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	12.6	14.0	13.7	12.9	13.0	13.2	13.3	14.9	14.7	14.4	12.1	(35)	
		MCDBR	16.8	17.9	20.3	19.4	16.9	15.5	16.2	16.2	17.1	20.1	20.5	16.3	(36)	
	5% WB	MCWBR	10.9	11.4	11.7	10.2	8.0	6.8	6.7	7.0	7.5	10.3	12.0	10.8	(37)	
		MCWBR	15.7	17.0	18.6	17.0	14.5	13.4	13.3	13.2	14.5	16.7	18.8	15.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.268	0.273	0.297	0.328	0.356	0.371	0.389	0.384	0.351	0.313	0.273	0.263	(40)	
		taud	2.500	2.502	2.466	2.396	2.401	2.389	2.381	2.377	2.426	2.511	2.540	2.508	(41)	
	Ebn at Noon	Ebn at Noon	902	952	958	943	919	901	881	875	885	889	891	879	(42)	
		Edh at Noon	72	84	98	114	116	118	118	115	102	83	69	66	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.20	3.11	4.14	5.18	6.02	6.93	6.89	5.95	4.81	3.41	2.41	1.90	(44)	
	RadStd	RadStd	0.19	0.19	0.36	0.27	0.43	0.46	0.24	0.29	0.29	0.35	0.17	0.15	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page